

Deva Sai Kumar Bheesetti

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SUMMARY

Software Engineer with 4+ years of experience building backend, full-stack, mobile, and AI-enabled applications using TypeScript, Python, NestJS, React, React Native, AWS, PostgreSQL, and MongoDB. Experienced in shipping production APIs, improving system performance, building CI/CD workflows, developing RAG applications, and owning features from design through deployment.

EDUCATION

University of Massachusetts Lowell

Master of Science in Computer Science, GPA: 3.94

Lowell, MA

Aug 2023 - Aug 2025

- **Coursework:** Machine Learning, Natural Language Processing, Algorithms, Human-AI Interaction

K L University

Bachelor of Technology in Electronics and Communication Engineering, GPA: 3.52

India

2016 - 2020

WORK EXPERIENCE

OurFreedom.AI

Full-Stack Software Engineer

New York, NY, Remote

Jan 2025 - Present

- Built and shipped production features across a NestJS, TypeScript, MongoDB, React Native, and AWS stack, owning backend APIs, mobile screens, Android release workflows, and deployment support.
- Designed a social graph system for follow, block, private profile, and follow request flows using REST APIs, dependency-injected services, cursor pagination, and MongoDB aggregation.
- Reduced profile and social graph load time by 60% by replacing repeated database calls with optimized aggregation pipelines, targeted projections, and joined response shapes.
- Implemented reporting and moderation workflows for posts, profiles, and direct messages with validation, self-report prevention, admin logging, and safe feature gating for compliance risks.
- Improved release reliability by standardizing dev, UAT, and production deployments with Docker, AWS ECS, GitHub Actions, Keybase-managed environment files, and structured service logging.

University of Massachusetts Lowell, Department of Computer Science

Graduate Research Assistant, AI and Healthcare

Lowell, MA

Jan 2024 - Aug 2025

- Built an NLP pipeline over 2,636 de-identified clinical notes to predict 15 Social Determinants of Health categories using PyTorch, Hugging Face, Clinical Longformer, DeepSeek-R1, and LLaMA 3.2.
- Improved model evaluation with ordinal classification, regression baselines, entropy-based active learning, Captum Integrated Gradients, and LIME, reaching 92.6% within-one ordinal accuracy.

University of Massachusetts Lowell, Manning School of Business

Graduate Assistant, Software Development and Data Automation

Lowell, MA

May 2024 - Aug 2025

- Automated reporting, grading, advising, and event operations using Python, JavaScript, Google Apps Script, Docker, Selenium, Tableau, and LLM-based data workflows, reducing manual workload by 70%.

Infosys Ltd.

Senior Software Engineer, Module Lead, Client: Verizon

Hyderabad, India

Aug 2022 - Aug 2023

Software Engineer, Client: Verizon

Dec 2020 - Aug 2022

- Owned the Publish Module for Verizon's offer management platform, supporting cross-team offer creation, validation, migration, and production operations for enterprise marketing workflows.
- Built REST API integrations between a ReactJS front end and Pega Decisioning Hub, with Oracle SQL data synchronization, Postman testing, and Kibana logging for better operational visibility.
- Resolved a critical production outage within 1 hour after a platform upgrade by tracing rule-class conflicts, coordinating onshore and offshore engineers, and strengthening error handling utilities.
- Improved team throughput by 30% through rule-based automation, scheduler-driven workflows, reusable components, and mentorship for a 4-member cross-functional engineering team.

PROJECTS

- **VitaScan, AI Health Assistant:** Created a cross-platform health triage app with Next.js, React Native, Expo, NestJS, Supabase, PostgreSQL, pgvector, RAG, OpenAI, Gemini, row-level security, usage quotas, symptom intake, chat support, and personalized recommendation flows. [Live] [GitHub]
- **TrustOps Platform:** Built a full-stack trust and safety operations platform with multi-tenant RBAC, JWT auth, moderation queues, report assignment, internal notes, audit logs, and a Next.js dashboard using NestJS, PostgreSQL, Prisma, Docker, Vercel, Render, and Supabase. [Live] [GitHub]
- **Support RAG Evaluator:** Developed a RAG evaluation platform for support-document retrieval, grounded AI answers, citations, refusal behavior, query logs, eval runs, streaming chat, OpenAPI docs, and pgvector search using NestJS, Next.js, Prisma, PostgreSQL, Docker, and GitHub Actions. [Live] [GitHub]

SKILLS

Languages: TypeScript, JavaScript, Python, Java, SQL, C++, Bash

Backend and APIs: NestJS, Node.js, FastAPI, Flask, REST APIs, JWT, Prisma, OpenAPI, microservices, authentication, RBAC

Frontend and Mobile: ReactJS, Next.js, React Native, Expo, Zustand, HTML, CSS, responsive UI development

Cloud, DevOps and Data: AWS EC2, AWS ECS, S3, Docker, Kubernetes, GitHub Actions, Jenkins, PostgreSQL, MongoDB, MySQL, Oracle SQL, Firebase, Supabase, Redis, pgvector

AI and Automation: RAG, LangChain, Hugging Face, PyTorch, TensorFlow, scikit-learn, pandas, NumPy, Selenium, BeautifulSoup, Google Apps Script